

★ WEEK 1 · PART 2 · PRACTICE

Stylized Facts, Worked by Hand

Measure the Moments, Then Split Direction From Size

Fat tails, negative skew, and volatility clustering

Part 2's empirical regularities, with the arithmetic shown. We compute the skewness and kurtosis of a return sample - finding fat tails and a left skew - then split the autocorrelation of returns from that of squared returns, the split that says direction is unpredictable but volatility is not.

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PART 2 · APPLICATIONS

What Returns Actually Look Like

Case A: compute the four moments of a return sample and find fat tails and negative skew - the crash-prone shape real returns take. Case B: contrast the autocorrelation of returns, which is near zero, with that of squared returns, which persists - volatility clustering. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

Returns are fat-tailed and left-skewed

Extreme moves happen far more than a normal predicts, and losses cluster in the left tail - markets fall faster than they rise.

2

Direction is unpredictable, size is not

The autocorrelation of returns is near zero, but the autocorrelation of squared returns persists - volatility comes in waves.

3

These are robust regularities, not theorems

Fat tails, negative skew, volatility clustering, and the leverage effect show up across markets and eras.

Everything here is a calculator exercise: one set of moments, one autocorrelation split.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a return sample.
- Compute moments, or autocorrelations.
- Then read the regularity.

What you should watch

- How excess kurtosis flags fat tails.
- How negative skew flags crashes.
- How squared returns reveal clustering.

The Shape of Returns

Fat tails and a left skew

A normal distribution has skewness zero and kurtosis three. Real returns don't. We take a return sample with one sharp crash, compute all four moments, and find a strongly negative skewness and a kurtosis well above three - fat tails. These two numbers capture why extreme losses are both more common and more severe than a bell curve suggests.

The sample

- Ten daily returns, in percent.
- Nine small moves, one -5.5% crash.
- Typical of a real return series.

What we'll do by hand

- Compute mean and variance.
- Compute skewness and kurtosis.
- Compare to the normal's 0 and 3.

The Third and Fourth Moments

Asymmetry and tail weight

$$\text{skew} = \frac{\overline{(r-\mu)^3}}{\sigma^3}, \quad \text{kurt} = \frac{\overline{(r-\mu)^4}}{\sigma^4} \quad (\text{normal: skew } 0, \text{ kurt } 3)$$

Skewness

- The average cubed deviation, standardized.
- Negative means a long left tail.
- Crashes outweigh melt-ups.

Kurtosis

- The average fourth-power deviation, standardized.
- Above three means fat tails.
- Extremes are more likely than normal.

The Four Moments

From the sample, by hand

moment	value	normal	reading
mean	-0.50%	-	slight loss
std dev	1.72%	-	the spread
skewness	-2.35	0	left-skewed
kurtosis	7.13	3	fat-tailed

The one crash dominates the higher moments: it drags the mean negative, pulls the skewness to -2.35, and lifts the kurtosis to 7.13 - more than double the normal's three. A single extreme observation reshapes the whole distribution.

Fat Tails and Negative Skew

What the numbers say

$\text{skew} = -2.35$ (< 0 : crashes), $\text{kurt} = 7.13 \Rightarrow \text{excess} = 4.13$ (> 0 : fat tails)

The excess kurtosis

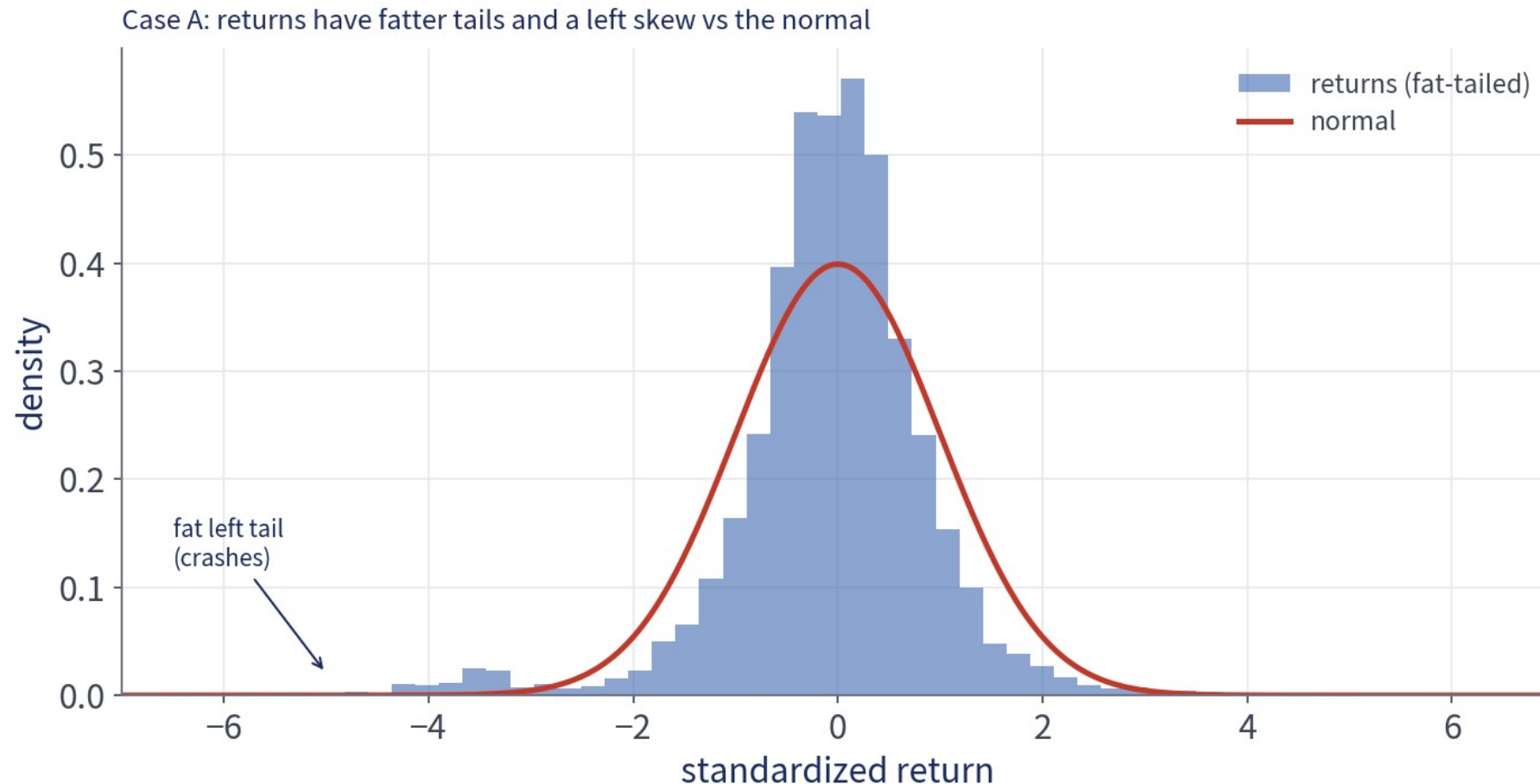
- Kurtosis 7.13 minus the normal's 3.
- An excess of 4.13, well above zero.
- Fat tails: extremes are common.

The negative skew

- A skewness of -2.35.
- The left tail is longer and heavier.
- Markets fall faster than they rise.

Fatter Tails, Longer Left

The return histogram sits taller in the center and, crucially, spills further into the tails than the red normal curve - especially on the left, where crashes live. That extra tail mass is the excess kurtosis; the leftward lean is the negative skew. A normal distribution would badly understate the risk of a large loss.



Why the Shape Matters

Risk lives in the tails

The danger of assuming normal

- A normal model prices tail risk far too cheaply.
- Six-sigma days should never happen - yet they do.
- Value-at-risk built on normality fails.

How heavy the tails are

- The tail follows a power law, not an exponential.
- A tail index around three to four is typical.
- The mean exists; higher moments barely do.

Direction vs Size

The autocorrelation split

Here is the course's most important split. Take a return series that runs calm, bursts into turbulence, then calms again. The autocorrelation of the returns themselves is near zero - you can't predict tomorrow's direction. But the autocorrelation of the squared returns is strongly positive - big moves follow big moves. Direction is unpredictable; size is not.

The series

- Twelve returns: calm, a burst, calm.
- The burst mixes signs.
- But the magnitudes cluster.

What we'll do by hand

- Compute the lag-1 autocorrelation of returns.
- Then of the squared returns.
- See the two diverge.

The Autocorrelation Function

Does the past predict the future?

$$\rho(k) = \frac{\sum_t (r_t - \mu)(r_{t-k} - \mu)}{\sum_t (r_t - \mu)^2}; \quad \text{band} = \pm \frac{1.96}{\sqrt{n}} \text{ (noise if inside)}$$

What it measures

- The correlation of the series with its own past.
- At each lag k , a single number.
- Inside the band means indistinguishable from noise.

The two versions

- On returns: tests direction.
- On squared returns: tests size.
- They tell very different stories.

The Split, Computed

Returns vs squared returns

$$\rho_r(1) = -0.15 \approx 0 \text{ (direction: no memory), } \rho_{r^2}(1) = +0.60 > 0 \text{ (size: clusters)}$$

Returns: no memory

- The lag-1 autocorrelation is about -0.15.
- Inside the noise band - effectively zero.
- Tomorrow's direction is unpredictable.

Squared: strong memory

- The lag-1 autocorrelation is about +0.60.
- Well outside the band.
- Big moves cluster: volatility persists.

The Key Idea

Returns unpredictable, volatility is not

returns unpredictable, volatility is not : $\rho_r(k) \approx 0$ but $\rho_{r^2}(k) > 0$ decaying slowly

Why they differ

- The signs of the burst cancel in returns.
- But squaring makes every big move positive.
- So the clustering survives in the squares.

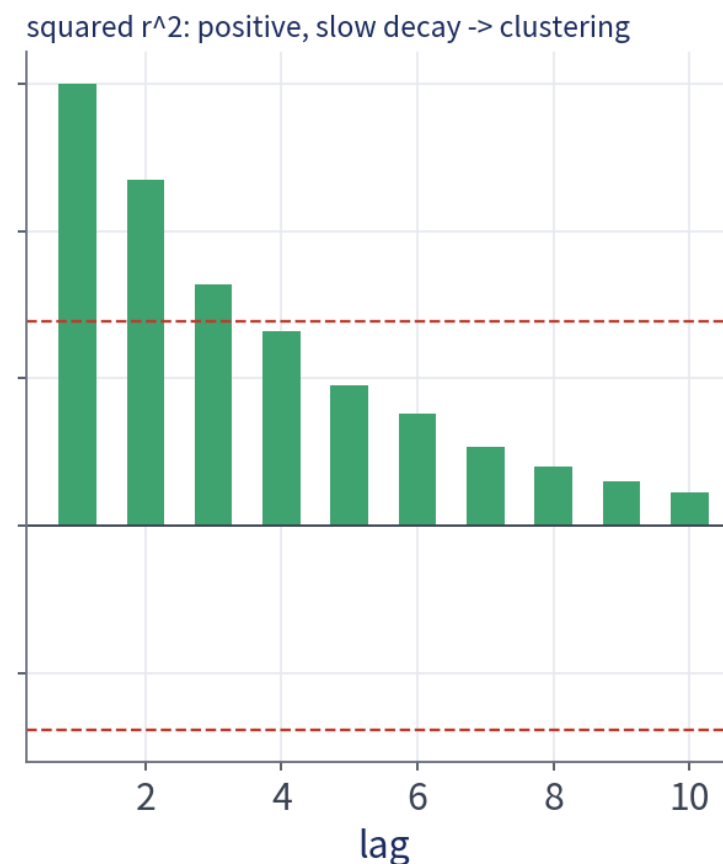
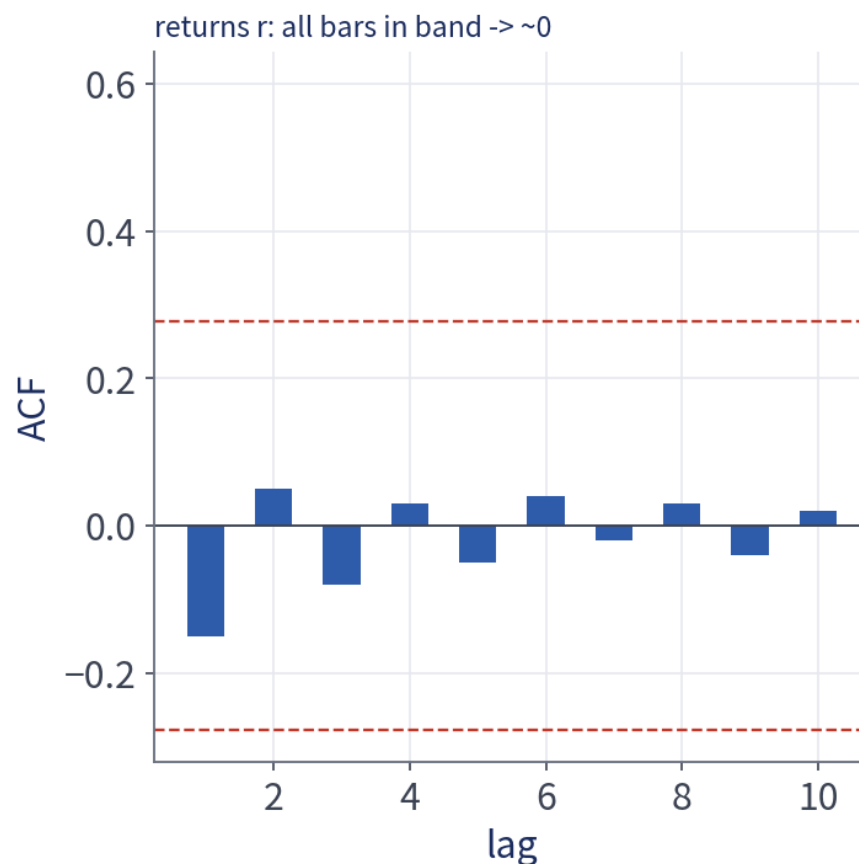
The leverage effect

- Volatility rises more after falls than rises.
- Bad news bites harder than good news.
- A related, direction-linked asymmetry.

Two Autocorrelations, Two Verdicts

Left, the autocorrelation of returns: every bar sits inside the significance band, so direction is unpredictable. Right, the autocorrelation of squared returns: strongly positive and slow to decay, so volatility clusters. This single contrast - flat on the left, persistent on the right - is the engine behind every volatility model to come.

Case B: no memory of direction, strong memory of size



What the Split Enables

And what fades with horizon

Why it's the key idea

- You can't forecast direction - markets are efficient.
- But you can forecast volatility - it clusters.
- Every GARCH model exploits exactly this.

Aggregational Gaussianity

- Daily returns are sharply non-normal.
- But monthly or yearly returns look more normal.
- Averaging washes out the fat tails.

The Stylized-Facts Checklist

What every return series shows

These regularities recur across markets and decades. Returns are fat-tailed and negatively skewed, so tail risk dwarfs what a normal predicts. Their autocorrelation is near zero - direction is unpredictable - but the autocorrelation of squared returns persists, so volatility clusters and can be forecast. Bad news raises volatility more than good news, and the non-normality fades as the horizon lengthens. Together these facts set the agenda for the whole course.

The moments (Case A)

- Skewness -2.35, kurtosis 7.13.
- Excess kurtosis 4.13 - fat tails.
- Negative skew - crashes.

The split (Case B)

- Returns autocorrelation about -0.15.
- Squared about +0.60 - clustering.
- Direction unpredictable, size not.

Reproduce the Moments and the Split

Direct Claude Code to check your arithmetic

GOAL Compute the four moments, then the autocorrelation of returns vs squared returns.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - moments:

- * for returns `[0.5,-0.4,0.6,-0.5,0.3,-0.6,0.4,-0.3,0.5,-5.5]`, print mean, sd, skewness, kurtosis;
- * report the excess kurtosis.

CASE B - the split:

- * for a calm-burst-calm series, print the lag-1 autocorrelation of the returns and of the squares;
- * confirm returns ~ 0 but squares > 0 .

Then plot the return histogram and the two ACFs.

Check against the slides

- skewness -2.35, kurtosis 7.13.
- excess kurtosis 4.13.
- returns lag-1 ~ -0.15 .
- squared lag-1 $\sim +0.60$.

Deliverable: `part2_practice.py` whose printed numbers match the moments and the autocorrelation split.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does excess kurtosis above zero mean?

**Fat tails - extremes
are more likely than
normal**

Q2 What does negative skewness indicate?

**A longer left tail -
crashes outweigh
melt-ups**

Q3 Why square the returns before the ACF?

**Signs cancel;
squaring reveals
volatility clustering**

All three come from the moment calculations and the autocorrelation split.

Three Pitfalls to Avoid

Where stylized facts trip people

Common mistakes

- "Returns are normally distributed."
- "Zero autocorrelation means nothing is predictable."
- "Skewness and kurtosis are minor details."

The reality

- They're fat-tailed and skewed - tails matter most.
- Direction isn't, but volatility is predictable.
- The higher moments drive tail risk.

Two Calculations to Remember

Part 2 applications in one screen

1

The moments reveal fat tails and skew

The return sample had a skewness of -2.35 and a kurtosis of 7.13 - an excess kurtosis of 4.13 over the normal's three. Fat tails and a heavy left tail: extreme losses are both more frequent and more severe than a bell curve would ever predict.

2

Direction is unpredictable, size is not

The lag-1 autocorrelation of returns was about -0.15, inside the noise band, so tomorrow's direction can't be forecast. But for squared returns it was about +0.60 - big moves follow big moves. Volatility clusters, and that is forecastable.

3

These facts set the course's agenda

Fat tails demand non-normal risk models; volatility clustering motivates GARCH; the leverage effect and aggregational Gaussianity refine both. Every later week answers a stylized fact seen here.

Next — Part 3 practice: market efficiency and the random walk, why direction resists prediction.